

王绍权 Shaoquan Wang

CTO / 技术副总裁 / 研发总监

Chief Technology Officer / VP of Technology / R&D Director

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01 职业概述 Professional Profile

挪威科学技术大学（NTNU）博士，杭州第四相物质科技有限公司共同创始人，曾任挪威 UBIQ 航空航天科技公司 R&D 工程师、杭州罗达氢能装备发展有限公司 CTO、中国民航大学副教授。兼具海外联合研发与国内技术创业经历，工作覆盖先进制造、航空复合材料与工程系统开发。

PhD, Norwegian University of Science and Technology (NTNU); co-founder of Hangzhou Fourth Phase Matter Technology Co., Ltd and former R&D Engineer at UBIQ Aerospace, Norway. CTO of Hangzhou Luoda Hydrogen Equipment and Technology Co., Ltd; Associate Professor of Civil Aviation University of China. Experience spans international R&D collaboration, technology entrepreneurship, advanced manufacturing, aerospace composites and engineering systems.

核心领域：电弧金属增材制造装备与工艺；航空复合材料及结构健康监测；无人机智能除冰；固态储氢系统与热管理。

Core expertise: wire arc additive manufacturing (WAAM) equipment and processes; aerospace composites and structural health monitoring; intelligent UAV icing protection; solid-state hydrogen storage and thermal management.

科研成果：第一作者 SCI 论文 7 篇；第一发明人专利记录 12 项。

Research output: 7 first-author SCI publications and 12 patent records as first-named inventor.

02 人才与荣誉 Honours and Recognition

浙江省海外引才千人计划青年人才

Young Talent, Zhejiang Province Overseas Thousand Talents Programme

杭州市 C 类人才

Hangzhou Category C Talent

ASRTU 优秀青年科研奖

ASRTU Outstanding Young Researcher Award

03 海外工作与项目经历 International Experience

挪威 UBIQ 航空航天科技公司 | 复合材料创新中心

UBIQ Aerospace, Norway | Composites Innovation Center

研发工程师 2021—2023

R&D Engineer, 2021–Jun 2023

- 参与无人机旋翼自主除冰系统研发，合作方包括 UBIQ Aerospace、Maritime Robot；挪威研究理事会项目，编号 296228。

Contributed to “Autonomous icing protection solution for unmanned aircraft rotors,” with UBIQ Aerospace and Maritime Robot; Research Council of Norway project 296228.

- 参与螺旋桨智能除冰飞行安全系统研发；合作方包括 UBIQ Aerospace、捷克 Mejlík Propellers；项目编号 318000。

Contributed to an intelligent propeller icing protection system with UBIQ Aerospace and Mejlík Propellers, Czech Republic; project 318000.

- 参与 Integrator UAV 智能除冰飞行安全系统商业项目，涉及 Insitu Defence、波音、洛克希德马丁、挪威国防部。

Contributed to the Integrator UAV intelligent icing protection commercial project involving Insitu Defence/ Boeing /Lockheed Martin and the Norwegian Ministry of Defence.

04 海外科研与项目经历 International Research Projects

挪威科学技术大学 (NTNU) Norwegian University of Science and Technology

工业工程系：研究员 / 博士研究，2018—2021。

Department of Industrial Engineering: Researcher / Doctoral Research, 2018–2021.

- 复合材料全寿命监测项目 (2018)：项目主持人；挪威研究理事会、NTNU 自由探索类研究，项目编号 177222。

复合材料实验室：研究员 / 技术顾问，2021—2022。

Composites Laboratory: Researcher / Technical Consultant, 2021–2022;

Project lead, full-life monitoring of composites (2018): investigator-led research at NTNU supported by the Research Council of Norway/ Norges forskningsråd; project 177222.

- THOR 热塑性储氢罐优化与回收项目 (2019—2023)：子项目负责人，涉及先进制造技术、装备及安全系统开发；European commission (欧盟委员会), IPN (研究与创新关键项目计划), 欧盟: Fuel Cells and Hydrogen 2 Joint Undertaking under grant agreement, 项目编号 826262。

Subproject lead, “THOR: Thermoplastic Hydrogen tanks Optimised and Recyclable” (2019–2023), covering advanced manufacturing technology, equipment and safety system development; European commission, IPN, EU Horizon: Fuel Cells and Hydrogen 2 Joint Undertaking under grant agreement, project 826262.

- H₂Enhance 德国与挪威先进氢能储运技术项目 (2021—2022)：挪威共同申请人。

Norwegian co-applicant, “H₂Enhance: Germany and Norway” (2021–2022), focusing on advanced hydrogen storage and transport technologies.

05 国内创业与先进制造 Technology Entrepreneurship in Mainland China

杭州第四相物质科技有限公司 | 共同创始人

Hangzhou Fourth Phase Matter Technology | Co-founder

2024 年至今。聚焦电弧金属增材制造、金属增减材装备及工程应用验证。

2024–Present. Focus on WAAM, hybrid metal additive/subtractive manufacturing equipment and engineering application validation.

职责：技术路线决策、核心研发、设备交付验收。

Responsibilities: technology strategy, core R&D, equipment delivery and acceptance testing.

- 国防重点专项相关构件的电弧金属增材技术、工装与工艺研究（2024—2025 年）。

WAAM technology, tooling and process research for main structural components in a key national defence programme (2024–2025).

- 国防工程相关金属增材设备研发与应用验证一期，涉及电弧金属增减材设备研发及技术保障（2024—2025 年）。

Phase I development and application validation of metal additive manufacturing equipment for a national defence engineering programme, including arc-based additive/subtractive equipment R&D and technical support (2024–2025).

06 其他国内任职与项目 Other Appointments and Projects

中国民航大学航空工程学院 | 副教授 | 2023—2024

Civil Aviation University of China | Associate Professor | 2023–2024

杭州洛达氢能装备有限公司 | CTO | Since 2023 年

Hangzhou Luoda Hydrogen Energy Equipment | CTO | Since 2023

技术方向：固态储氢装备、绿色储能系统及储氢热电联供工程。

Technical focus: solid-state hydrogen storage equipment, green energy storage and hydrogen-based combined heat and power (CHP) systems.

相关储氢工程项目 Related hydrogen projects | 2023 年起 / Since 2023

- 杭州施福宁能源固态储氢绿色储能系统（2023 年）。

Solid-state hydrogen-based green energy storage system for Hangzhou Shifuning Energy (2023).

- 杭州萧山区绿色产业新城零碳固态储氢热电联供储能示范项目。

Zero-carbon solid-state hydrogen storage and CHP demonstration project in the Green Industry New City, Xiaoshan, Hangzhou.

- 浙江交投中碳环境科技有限公司次坞高速公路服务区固态储氢热电联供系统。

Solid-state hydrogen storage and CHP system at Ciwu Expressway Service Area for Zhejiang Communications Investment-Zhongtan Environmental Technology.

07 教育经历 Education

2018—2021 博士 | 挪威科学技术大学 (NTNU)

研究方向: 复合材料制造与结构健康监测

2018—2021 PhD | Norwegian University of Science and Technology

Research: Composite manufacturing and structure health monitoring

2010—2013 硕士 | 哈尔滨工业大学 | 空间材料与加工

2010—2013 Master's Degree | Harbin Institute of Technology

Space Materials and Processing

2006—2010 本科 | 中国石油大学 (北京) | 材料科学与工程

2006—2010 Bachelor's Degree | China University of Petroleum (Beijing)

Materials Science and Engineering

08 第一作者论文 First Author Publications

共 7 篇 SCI 论文。7 SCI publications, all as first author.

1. 2022 | Materials & Design

Integration of optical fibre sensors by material extrusion 3-D printing – The effect of bottom interlayer thickness
通过材料挤出式 3D 打印集成光纤传感器——底部层间厚度的影响

2. 2021 | Sensors

Integration of optical fiber Comparison of DOFS Attachment Methods for Time- Dependent Strain Sensing
光纤集成与用于时变应变传感的分布式光纤传感器 (DOFS) 固定方法比较

3. 2020 | Additive Manufacturing

Novel in-situ residual strain measurements in additive manufacturing specimens by using the Optical Backscatter Reflectometry

利用光学背向散射反射测量技术对增材制造试样进行新型原位残余应变测量

4. 2020 | Optics express

Accurate non-linear calculation model for decoupling thermal and mechanical loading effects in the OBR measurements

用于解耦 OBR 测量中热载荷与机械载荷效应的精确非线性计算模型

5. 2019 | Composites Part A: Applied Science and Manufacturing

Comparison of the mechanical deterioration behavior of C/BMI composite under hydro-thermal or vacuum thermal cycling

C/BMI 复合材料在湿热或真空热循环下的力学性能退化行为比较

6. 2018 | Materials & Design

Thermal ageing effects on mechanical properties and barely visible impact damage behavior of a carbon fiber reinforced bismaleimide composite

热老化对碳纤维增强双马来酰亚胺复合材料力学性能及低可见度冲击损伤行为的影响

7. 2018 | Polymer composites

Low-velocity impact behavior of a carbon/bismaleimide composite proposed for supersonic flight simulation after hygrothermal cycling

湿热循环后用于超声速飞行模拟的碳纤维 / 双马来酰亚胺复合材料低速冲击行为

09 第一发明人专利 Patents as First Named Inventor

第一发明人：王绍权；共 12 项，包括 4 项发明专利、8 项实用新型。

First-named inventor: Shaoquan Wang. 12 records: 4 inventions and 8 utility models.

专利名称 Patent title	申请号与类型 Application and type	状态 Status
基于低温安全固态储氢材料的生氢发电系统及控制方法 Hydrogen generation and power system based on low-temperature safe solid-state hydrogen storage materials, and control method	CN202510055063.5 发明专利 / Invention	授权 Granted
一种固态储氢热管理系统及方法 Solid-state hydrogen storage thermal management system and method	CN202510054968.0 发明专利 / Invention	授权 Granted
一种新型固态储氢热管理系统及方法 Novel solid-state hydrogen storage thermal management system and method	CN202510054969.5 发明专利 / Invention	授权 Granted
一种液氢存储球罐增材制造装置及方法 Additive manufacturing apparatus and method for spherical liquid hydrogen storage tanks	CN202411185547.3 发明专利 / Invention	待授权 Pending
一种固态储氢与燃料电池的换热系统 Heat exchange system for solid-state hydrogen storage and fuel cells	CN202422529753.3 实用新型 / Utility model	授权 Granted
一种实验用固态储氢设备 Experimental solid-state hydrogen storage equipment	CN202422533414.2 实用新型 / Utility model	授权 Granted
一种固态储氢设备 Solid-state hydrogen storage equipment	CN202422538339.9 实用新型 / Utility model	授权 Granted
一种固态储氢罐 Solid-state hydrogen storage tank	CN202422468792.7 实用新型 / Utility model	授权 Granted
一种储氢装置 Hydrogen storage device	CN202422411699.2 实用新型 / Utility model	授权 Granted
一种便于存取的储氢装置 Hydrogen storage device for convenient storage and retrieval	CN202422411459.2 实用新型 / Utility model	授权 Granted
一种储氢装置用冷却装置及储氢装置 Cooling device for a hydrogen storage device, and hydrogen storage device	CN202422409319.1 实用新型 / Utility model	授权 Granted
固态储氢罐及车载固态储氢系统 Solid-state hydrogen storage tank and vehicle-mounted solid-state hydrogen storage system	CN202421931644.8 实用新型 / Utility model	授权 Granted